Hetkumar Patel

Applied AI & Agentic Systems
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Summary

Applied AI software engineer who designs, builds, and ships **production-grade, LLM-powered agentic systems** from prototype to deployment. Hands-on across the full LLM stack, frontier models (Claude, GPT, Gemini), **RAG, multi-step planning agents (ReAct / Plan-and-Execute), MCP and multi-agent orchestration** with a strong focus on **rigorous evaluation frameworks** that measure agent accuracy, safety and latency beyond trial-and-error. Writes clean, testable, observable **Python** and flexes into frontend (Next.js / React) when the problem demands it. Co-author of three AI/ML papers; comfortable working directly with stakeholders to translate ambiguous business problems into reliable, auditable solutions, including in regulated domains like healthcare.

Experience

Software Engineer: AI Agents — **Jan. 2026 – Present**
Wedge (Y Combinator S25) — *Remote*

- Build **production LLM agents for healthcare workflows** automating referral intake, follow-up and payer communication for clinics in a **regulated, sensitive environment**.
- Shipped an **Automated Referral System end-to-end**: fax ingestion + LLM-assisted parsing (pdfplumber + Groq), structured records in Supabase, and an agent + UI for triage, with Playwright automation for payer portals.
- Built an **outbound voice agent** on LiveKit (Deepgram STT + ElevenLabs TTS + OpenAI reasoning) that calls insurance payers for verification and prior-authorization follow-up, tuned for low-latency turn-taking on live calls.
- Implemented Twilio + APScheduler SMS/voice follow-up and a knowledge layer feeding the agents, in clean **Python (FastAPI, Pydantic)**.

Software Developer: Agentic AI Systems — **Feb. 2026 – Present**
Moorcheh.ai by Edge AI Innovations — *Toronto, ON · Remote*

- Built and delivered **Memanto**, an **AI memory-layer agent**, end-to-end on Moorcheh's architecture; Co-authored an arXiv paper on its typed semantic memory and information-theoretic retrieval for long-horizon agents.
- Engineered the core memory pipeline on **RAG semantic search**, persisting and retrieving context across sessions, and exposed memory as a first-class tool via **MCP** integrated with multi-agent workflows.
- Built the **evaluation pipeline** (LongMem / LoCoMo benchmarks) and iterated on retrieval, latency and reliability for a production agentic system; shipped the Next.js landing + MDX docs sites.

Gen AI Associate: LLM Evaluation — **Jan. 2026 – Mar. 2026**
Innodata Inc. — *Toronto, ON · Remote*

- Ran **structured LLM evaluation workflows** across locality, freshness, completeness and on-point accuracy, applying A/B-style testing and human-in-the-loop review to measure and improve model output quality.
- Contributed evaluation insights to **large-scale model-improvement pipelines** across multiple model versions.

Artificial Intelligence Researcher (Part-Time) — **Sep. 2024 – Aug. 2025**
Sheridan College, Centre for Applied AI — *Oakville, ON · Hybrid*

- Built an end-to-end **RAG pipeline on LangChain** with vector indexing for KPI extraction from unstructured fleet documents, and **fine-tuned LLMs (LoRA / PEFT)** on domain corpora with documented evaluation benchmarks.
- Deployed **agentic workflows with function calling, tool orchestration and automated quality checks**; co-authored **two peer-reviewed papers** presented at international conferences and mentored junior researchers.

AI/ML Software Developer (CO-OP) — **May 2024 – Sep. 2024**
Naryant, Centre for Applied AI — *Oakville, ON · Hybrid*

- Owned the full ML lifecycle for a transportation mode detection system in **Python, Scikit-learn, TensorFlow** (**92% accuracy**), and built a **CTGAN synthetic-data pipeline** for SUMO mobility simulations.

IIoT Developer (CO-OP) — **May 2023 – Feb. 2024**
Magna International — *Milton, ON · On-site*

- Delivered real-time monitoring (Ignition + Python) that cut troubleshooting time by **85%** and improved monitoring accuracy by **97%** across production machinery.

Projects

ContextIQ: Production RAG System | *FastAPI, LangChain, Pinecone, pgvector, React, Docker* | **GitHub** **Mar. 2025**

- End-to-end RAG application with document ingestion, hybrid vector search (Pinecone + pgvector), re-ranking and query rewriting; FastAPI backend with groundedness scoring and structured-output validation for auditability, containerized with Docker.

Restro: Multilingual Voice Agent | *LiveKit, Claude, Deepgram, Cartesia, Twilio SIP* **2026**

- Real-time multilingual restaurant voice agent (English, Hindi, Gujarati, Punjabi) with full pipeline: SIP ingress, VAD, streaming STT, LLM reasoning, low-latency TTS and turn-taking logic.

Technical Skills

Agentic & LLM: Agent design (ReAct, Plan-and-Execute), MCP, multi-agent orchestration (LangChain, LangGraph, CrewAI, AutoGen, Semantic Kernel), tool/function calling, RAG, semantic memory, prompt engineering, guardrails, LoRA/PEFT fine-tuning

LLM APIs & Frameworks: Anthropic Claude, OpenAI, Google Gemini, Vertex AI, Groq; Hugging Face, PyTorch, TensorFlow, Scikit-learn, llama.cpp

Evaluation & MLOps: Evaluation frameworks (accuracy/safety/latency), A/B testing, human-in-the-loop, groundedness scoring, LongMem/LoCoMo benchmarks, pytest, MLflow, model versioning, CI/CD, GitHub Actions, Docker, Kubernetes

Backend & Frontend: Python (FastAPI, asyncio, Pydantic), Flask, REST APIs; TypeScript, Next.js, React, Tailwind; C#, ASP.NET

Vector & Data: Pinecone, pgvector, Chroma, Milvus, Redis, hybrid search, re-ranking; SQL, MongoDB, Supabase, Pandas, Spark, Databricks

Voice & Cloud: LiveKit, Deepgram, ElevenLabs, Twilio SIP, Silero VAD; AWS, Azure, GCP, Terraform, Vercel

Research, Education & Certifications

Publications: (1) Memanto: Typed Semantic Memory with Information-Theoretic Retrieval for Long-Horizon Agents (arXiv); (2) Integrating Knowledge & Data-Driven Methods to Generate Synthetic Mobility Data (IEEE); (3) Topic Modeling Enhancement using LLM-Generated Summaries (IEEE)

Education: Honours Bachelor of Computer Science (Data Analytics), Sheridan College – GPA 3.7, Awarded Honours (2025)

Certifications: Prompt Engineering & Programming with OpenAI and Machine Learning I — Columbia (2025); OCI Generative AI Professional, Data Science & AI Foundations — Oracle (2025); Generative AI Fundamentals — Databricks; Building GenAI Applications — MongoDB; AWS Educate: Intro to Generative AI; GitHub Foundations — Microsoft (2025)